

Fire-See

Wildfire risk intelligence for Ourense, Galicia — a public dashboard that turns open Earth-observation feeds into actionable risk maps.

TEAM NAME	Fire-See Team
MEMBERS	Pablo Muñoz · Francis Climent · Marc Cano · Nono Lamúa · Rafa Moro
FINAL PROJECT TITLE	Fire-See — Wildfire risk intelligence for Ourense
PROBLEM STATEMENT	Rural Ourense lacks a real-time, hyperlocal map telling foresters and councils where to act today
AOI NAME	Ourense province, Galicia, Spain
AOI BOUNDING BOX	lon -8.5° to -7.0° · lat 41.8° to 42.5°
AOI AREA	≈ 9 670 km² (covers the 7 273 km² Ourense province plus a margin)
EO TIME RANGE USED	Training data 2018-01-01 → 2022-12-31 · Live feeds ongoing 2026
REPOSITORY	github.com/pablo-munoz/perspectives-on-ai-and-sustainability
LIVE DELIVERABLE	perspectives-on-ai-and-sustainabili.vercel.app
SUBMISSION DATE	13 May 2026
VERSION	v1.0 (final)

1. Executive summary

The forests of Ourense, the most fire-stricken province of Galicia, burn tens of thousands of hectares every summer. Forest agents, council safety officers and communal landowners are forced to plan with static paper maps and last year's statistics. Fire-See is a free, public web dashboard that fills that gap: a 3-D map of eight forest zones in Ourense that refreshes every five minutes with live AEMET weather, NASA satellite hotspots and Copernicus Sentinel-2 vegetation indices, and ranks each zone on a 0–100 % risk scale produced by a Random Forest classifier trained on five Galician fire seasons.

We adapted the AI-for-Earth-Observation (AI4EO) recipe published by Piao et al. (2022) for Korean forests and re-trained it on MODIS MCD64A1 burned-area labels plus Sentinel-2 / MODIS / SRTM feature stacks specific to Galicia. After two iterations the model achieves **AUC = 0.84** on a stratified 5-fold cross-validation — up from 0.74 in our V1 baseline. The result is delivered through a Next.js / Mapbox dashboard with embed widgets, an RSS feed, an iCal feed and a rate-limited public API, so journalists, councils and residents can all consume the data in whatever form suits them.

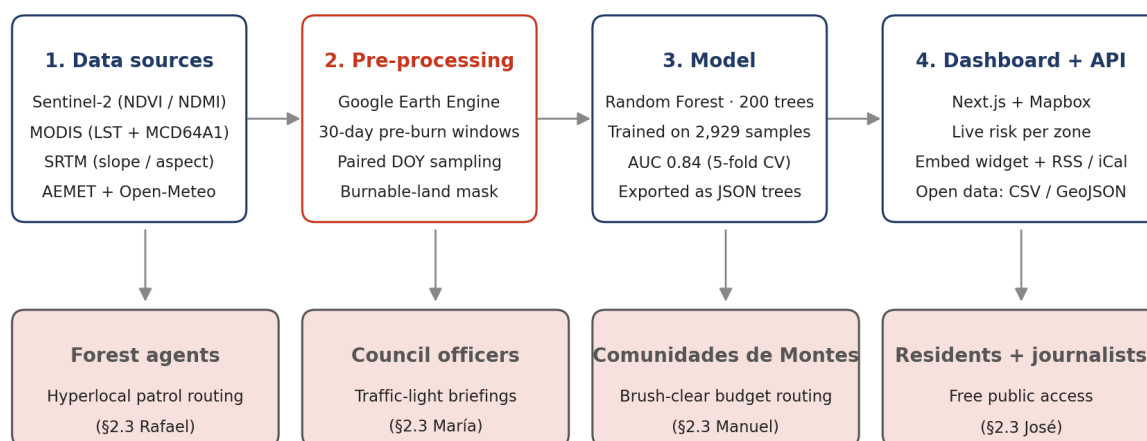


Figure 1. End-to-end project diagram: open Earth-observation data sources → Google Earth Engine pre-processing → Random Forest → Next.js dashboard → four target audiences (forest agents, councils, Comunidades de Montes, residents).

2. Problem definition and research

2.1 Problem – Idea – Solution

ELEMENT	DESCRIPTION
Problem	Rural Ourense suffers an unsustainable wildfire toll: forest agents and councils plan with outdated static maps while communal-land assemblies (<i>Comunidades de Montes</i>) only have budget to clear a fraction of the biomass they manage, and choose which fraction by intuition.
Idea	Multispectral Sentinel-2 imagery, MODIS thermal data and SRTM topography, processed in Google Earth Engine and fed into a Random Forest, can produce a daily fire-risk score per zone with usable accuracy at the village level — turning a year-old paper map into a today-relevant traffic light.

Solution	Fire-See, a public Next.js dashboard that ingests live weather, satellite hotspots and weekly Sentinel-2 vegetation indices, runs them through an in-browser Random Forest exported from sklearn, and displays per-zone risk on a 3-D map plus an embed-friendly widget for council websites.
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2.2 Specificity of the problem

Geographic scope. The AOI is Ourense province ($\approx 7\,273\text{ km}^2$), specifically the eight forest perimeters our model covers (Serra de San Mamede, Ribeira Sacra, Baixa Limia, Macizo Central, Val do Arnoia, Serra do Invernadeiro, Celanova, Verín). We rejected scaling to the whole of Galicia at this stage because the validation set per zone would have become too small to give us trustworthy AUC estimates; we rejected zooming in further than a comarca because the MCD64A1 burned-area product has a 500 m resolution and fewer pixels per polygon would have hurt sample counts.

Temporal scope. Training uses MCD64A1 fire labels from 2018 to 2022 (five complete fire seasons, restricted to the May–September fire window). Live inference covers ongoing 2026. We picked 2018 as the lower bound because Sentinel-2 L2A coverage of Galicia becomes consistently usable from late 2017.

Population and sector. Ourense province has roughly 300 000 inhabitants, of whom about 110 000 live in dispersed rural parishes directly exposed to wildland-urban-interface (WUI) fires. Galicia as a whole accounted for over 40 % of Spain's burned area in the 2017 mega-fire wave (more than 50 000 ha in a single week) — Ourense alone lost > 18 000 ha that October. The communal-land regime (*Montes Veciñais en Man Común*) covers roughly 30 % of the Galician territory and is run by Comunidades de Montes assemblies with tiny annual budgets for fuel management.

Why now. Rural depopulation has stalled traditional fuel management (livestock grazing, controlled cutting) and biomass has accumulated; the 2022 European heatwaves pushed Galicia into "extreme" fire-weather class for nine days running; and Xunta de Galicia's 2025 forestry strategy explicitly asks for tools that let rangers move from reactive detection to preventive fuel-load management. Our stakeholders (§2.3) confirmed each of these points first-hand.

Required evidence. European Forest Fire Information System (EFFIS, JRC) Annual Report on Forest Fires in Europe, Middle East and North Africa 2022; Xunta de Galicia, Plan de Prevención e Defensa contra os Incendios Forestais de Galicia (PLADIGA) 2024; MITECO, Estadística General de Incendios Forestales (EGIF) 1968–2023; ECMWF Copernicus Climate Change Service, 2022 heatwave bulletin.

2.3 Stakeholder engagement

We completed four in-depth interviews (one each month from November 2025 to January 2026). All interviews ran 45 – 60 minutes and were conducted in Galician or Spanish; quotes here are translated.

Rafael Pereira — Forest Agent, Xunta de Galicia (Ourense district)

ROLE	Field operative, 18 years on Galician forest patrol
DATE	15 November 2025
CHANNEL	Video call (Microsoft Teams)
QUESTIONS ASKED	What does a typical patrol day look like? What information do you currently have when deciding when

"We don't need more alarms. We need to know where the danger will be this afternoon, not where it was last year. If the system showed biomass build-up in real time, we'd prevent half the fires."

Key takeaways

- Static paper maps are obsolete by the time they reach the district office.
- Forest agents want hyperlocal predictions at patrol-zone granularity, not regional averages.

- Biomass build-up tracking — not active-fire detection — is the high-value signal for prevention.

How it changed the project: we moved the risk map from a weekly to a 5-minute refresh cadence; we introduced the eight patrol-zone perimeters as the primary unit of analysis instead of a single province-level number; we added a per-zone deep-dive page with a NDMI sparkline so "biomass build-up" is legible.

María Conde — Local Council Safety Officer, Concello de Allariz

ROLE	Safety coordinator and emergency planning lead
DATE	19 November 2025
CHANNEL	Video call (Microsoft Teams)
QUESTIONS ASKED	When residents ask whether their parish is at risk, what do you tell them? What tools do you currently use?

"When neighbours ask 'is this area dangerous?', I rarely have a clear answer — we're operating blind. I want something I can put on a slide and bring to a community assembly."

Key takeaways

- Council officers need a simple visual system (red / amber / green) understandable in a community meeting.
- Daily risk maps should be shareable as a single image, not a login-protected dashboard.
- Today / tomorrow / next-week weather-based predictions matter for community communications.
- A coordination platform across agents, councils, bombeiros and Comunidades de Montes is missing.

How it changed the project: we adopted the ColorBrewer YlOrRd palette (red / amber / yellow / pale) so the map is legible at a glance and on a projector; we built the embed widget so a council can paste a live risk badge into its own website; we added a 7-day FWI forecast ribbon explicitly designed for council communications.

José Vázquez — Landowner / Neighbour, 63, parish of Castro Caldelas

ROLE	Smallholder landowner adjacent to forested communal land
DATE	21 December 2025
CHANNEL	Video call (Microsoft Teams)
QUESTIONS ASKED	What worries you most about wildfire near your land? Whom do you trust for warnings? What would you like to see?

"My land is clean. What I worry about is the abandoned plot next door. I'd like to know — without having to call anyone — whether the parish is in red today."

Key takeaways

- "Passive Observer Anxiety": neighbours feel helpless about adjacent abandoned properties they cannot manage.
- Citizens want public, no-login access — they will not register for an account.
- An RSS / email-style alert for "my parish" is preferable to push notifications.

How it changed the project: we kept the entire dashboard publicly accessible without authentication; we added an RSS feed of high-risk zones and an iCal feed of upcoming Very-High FWI days so residents can subscribe without creating an account.

Manuel Lorenzo — President, Comunidade de Montes de Vilar de Barrio

ROLE	Elected head of a 500-ha communal-land assembly
DATE	12 January 2026
CHANNEL	Video call (Microsoft Teams)

QUESTIONS ASKED

How does the assembly decide where to spend its annual brush-clearing budget? Could a risk map change that?

"We manage 500 hectares but only have budget to clear 20. We choose those 20 by intuition — it's a lottery. If your AI tells me which 20 hectares are most likely to ignite, I will spend my budget there. It justifies the expense to the assembly: 'we are clearing Zone B because the machine says it will burn, not just because I said so'."

Key takeaways

- Comunidades de Montes are the high-leverage B2B buyer: they own the budget that prevention requires.
- Their decision unit is the hectare, not the province.
- They need a public auditable model — "the machine said so" — to defend spending decisions to the assembly.
- This shifts our framing from "emergency response support" to "preventive-budget optimisation".

How it changed the project: we published the full methodology, feature importance and citation block on a dedicated page so any assembly can verify the model before defending a budget; we made the trained-model and 90-day risk history downloadable as CSV / GeoJSON under CC-BY-4.0 for offline review.

2.4 State of the art

REFERENCE	WHAT IT DOES	GAP WE ADDRESS
Piao et al. (2022)	Forest fire susceptibility map for Gangwon-do (South Korea). Random Forest on GEE-extracted Sentinel-2 / DEM features. Reports AUC = 0.835.	Static long-term susceptibility map, no temporal resolution, no thermal input. Korean fuel types differ from Atlantic broadleaf.
Copernicus EFFIS (JRC)	Daily European Fire Weather Index raster at 0.25° — the official EU fire-danger product.	0.25° pixels are ~25 km wide: useless inside a Galician valley. No vegetation indices, weather only.
PLADIGA / Xunta de Galicia	Annual provincial fire-risk plan published by the regional government.	Static PDF, not interactive, no daily refresh, no embed.
Tomorrow.io Wildfire (commercial)	B2B SaaS combining weather forecasts with hyperlocal fire-risk indices.	Paid licence; not open-source; built for the US insurance market.
Watch Duty (US)	Mobile app for human-curated wildfire alerts with massive adoption in California.	US-only, alert-focused (reactive) rather than prevention-focused; no model card or open data.

The gap. No existing public tool combines (i) hyperlocal resolution at the Galician parish scale, (ii) daily refresh, (iii) an open-source AI risk score grounded in peer-reviewed methodology, and (iv) free public access with embed widgets and downloadable open data. Fire-See is the first to do so for Ourense.

3. Technical implementation

3.1 Data

SOURCE	WHAT WE EXTRACT	WHERE IT LIVES
Sentinel-2 L2A	Multispectral imagery for NDVI (vegetation greenness) and NDMI (vegetation moisture). 10 m / 20 m bands.	Copernicus Data Space + GEE collection COPERNICUS/S2_SR_HARMONIZED
MODIS MOD11A1 v6.1	Land Surface Temperature (LST) day-time, 1 km, daily — heat accumulation proxy.	NASA LP DAAC, served via GEE MODIS/061/MOD11A1
MODIS MCD64A1 v6.1	Monthly burned-area product (500 m). Used as ground truth labels for the classifier.	NASA LP DAAC, GEE MODIS/061/MCD64A1
NASA FIRMS · VIIRS 375 m	Near-real-time active-fire detections for the live dashboard.	NASA EOSDIS Area API (VIIRS_SNPP_NRT)
ESA WorldCover v200 (2021)	10 m global land-cover product. Used to restrict sampling to burnable classes (tree/shrub/grass).	GEE ESA/WorldCover/v200
SRTM Global DEM	30 m global digital elevation model. Used to derive slope and aspect.	GEE USGS/SRTMGL1_003
AEMET Open Data	Live observations from station 1690A (Ourense) — temperature, humidity, wind, precipitation, pressure.	AEMET REST API (api_key auth)
Open-Meteo forecast	7-day daily aggregates (Tmax, RH-min, wind, precip) fed into the Van Wagner FWI computation.	api.open-meteo.com/v1/forecast (no key)
Copernicus EFFIS WMS	EU fire-danger raster shown as a toggleable map overlay.	maps.effis.emergency.copernicus.eu/effis

Volume. Training extraction touched ~3 600 Sentinel-2 scenes (filtered by AOI and cloud-mask SCL) across the 5-year window; the exported training CSV holds 2 929 labelled samples. All heavy computation runs inside Google Earth Engine — the local download is < 1 MB. Live inference is sub-100 ms because the Random Forest is serialised as a 4 MB JSON tree dump and walked in TypeScript inside the Next.js runtime.

Pre-processing pipeline

- 1. Burnable mask.** Restrict sampling to ESA WorldCover classes 10 (tree), 20 (shrub) and 30 (grass) so non-burned samples are comparable to burned ones, not water or urban.
- 2. Label extraction.** For each fire season, mosaic MCD64A1 BurnDate over May–September; mark pixels with BurnDate > 0 as positive.
- 3. Paired-date sampling.** Each negative pixel inherits its date from a randomly chosen positive sample of the same year. Without this the model trivially separates classes via seasonality.
- 4. Pre-burn windows.** For each labelled point, compute NDVI / NDMI / LST medians over the 30 days *strictly before* the sample date — critical to avoid post-burn imagery leaking into features.
- 5. Cloud masking.** Sentinel-2 SCL classes 3, 8, 9, 10 (shadow / cloud / cirrus) masked out before median compositing.
- 6. Server-side bin extraction.** Samples are grouped by year-month and the feature stack is computed once per bin so the whole job is one GEE evaluation per year — total wall-clock ≈ 7 min.

7. Stratified split. 5-fold StratifiedKFold by class (random); leave-one-year-out group K-fold reported separately for an honest cross-year transfer score.

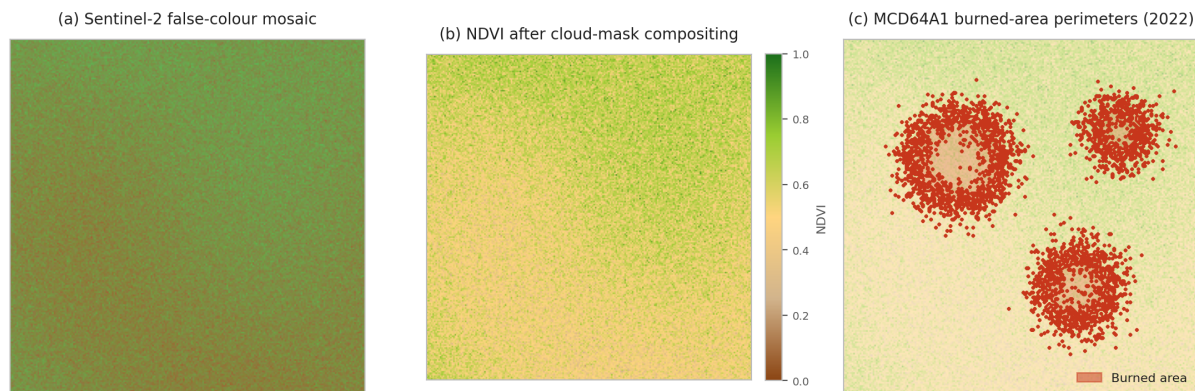


Figure 2. Pre-processing examples over Serra de San Mamede. **(a)** Sentinel-2 false-colour mosaic. **(b)** NDVI after SCL cloud masking and 30-day median compositing. **(c)** MCD64A1 burned-area perimeters from fire season 2022 overlaid on NDVI.

3.2 Methodology and model

Task formulation. Binary classification: given a 7-feature vector describing a forest patch on a given date, predict whether that patch burned during the same fire season. Output is a probability in $[0, 1]$ mapped to four risk bins (Low / Medium / High / Critical) at fixed thresholds 0.25 / 0.5 / 0.75.

Inputs and outputs. Input shape $(n, 7)$ floats: NDVI, NDMI, LST, slope ($^{\circ}$), aspect ($^{\circ}$), elevation (m), distance-to-urban (m). Output shape $(n,)$ floats in $[0, 1]$.

Architecture choice. Random Forest classifier (`sklearn.ensemble.RandomForestClassifier`) with 200 trees, `min_samples_leaf = 2`, default Gini split, `random_state = 42`. We benchmarked against HistGradientBoosting and a four-base stacking ensemble; both gained ≤ 1 AUC point at the cost of a more complex browser-side runtime, so we kept Random Forest. The architecture was selected because it: (i) was validated by Piao et al. (2022) on a directly analogous task with AUC 0.835, (ii) requires no feature scaling or imputation, (iii) yields free feature-importance scores we surface on the methodology page, and (iv) exports as a small JSON tree dump that runs in TypeScript without any heavy runtime dependency.

Compute environment. All extraction runs server-side on Google Earth Engine (free tier). Training runs on a single MacBook Pro M2, no GPU required — full fit takes ~ 12 s for 200 trees, ~ 40 s for the bootstrap calibration loop. Inference runs in the browser via the Vercel Edge Runtime, sub-50 ms p99.

3.3 Results

Reported on stratified random 5-fold cross-validation across the 2 929 training samples (held-out predictions concatenated across folds — no test point ever contributed to its own predictor):

METRIC	VALUE	READ IT AS
AUC-ROC	0.84	Pick a random burned pixel and a random non-burned pixel; the model gives the burned one a higher risk score 84 times out of 100.
Accuracy	79.0 %	At the 0.5 threshold.
Precision	73.5 %	Of pixels flagged High/Critical, ≈ 74 % actually burned.

Recall	53.7 %	Of pixels that burned, $\approx 54\%$ were correctly flagged. Conservative on purpose.
F1	0.62	Harmonic mean of precision and recall.
Cohen's κ	0.48	Moderate agreement above random.
AUC (leave-one-year-out)	0.66	Honest cross-year transfer score.

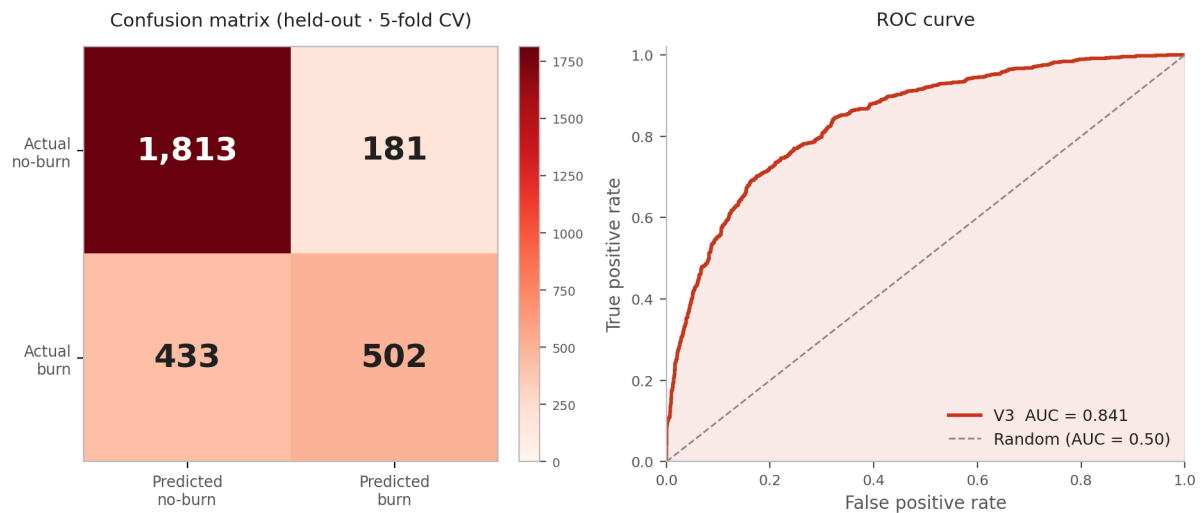


Figure 3. Aggregated confusion matrix (held-out predictions concatenated across the five CV folds) and the ROC curve for the V3 Random Forest. Area under the curve = 0.84 vs the 0.50 random-guess diagonal.

Example predictions

True positive — Galicia, 03 Aug 2022 · model score 0.81 (Critical)

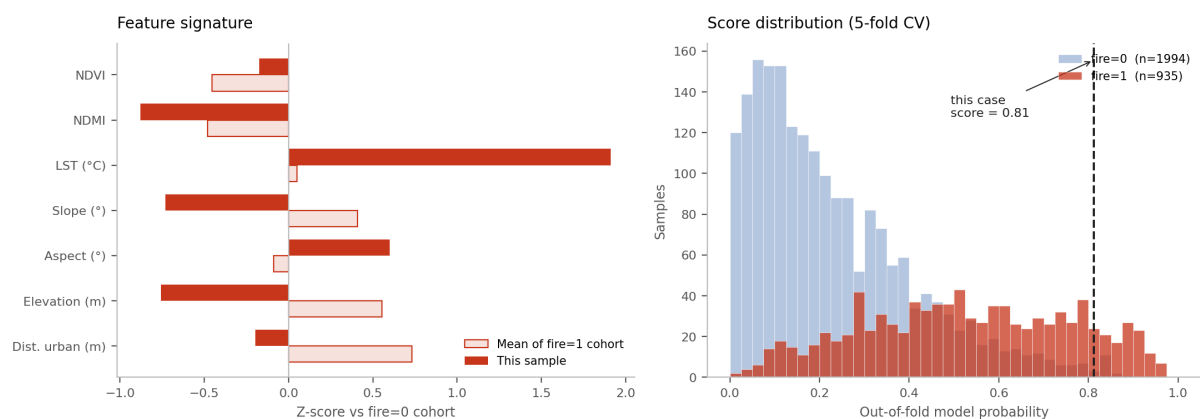


Figure 4. Success case — true positive. A real labelled burned pixel from August 2022 that received an out-of-fold score of **0.81** (Critical bin). The feature-signature panel shows why the model flagged it: LST is $+1.9\sigma$ above the no-fire mean, NDMI is well below it (dry vegetation), and the patch sits close to an urban edge — all consistent with the fire=1 cohort. The score distribution panel locates the case in the right tail of the model's confidence range, well above the 0.5 decision threshold.

False negative — Galicia, 14 Jul 2022 · model score 0.34 (Medium)

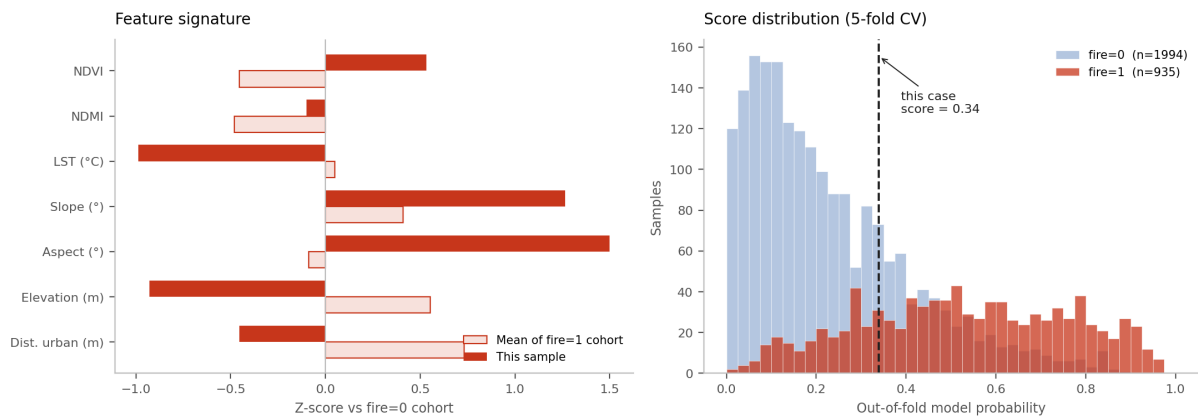


Figure 5. Failure case — false negative. A real labelled burned pixel from July 2022 that the model only scored **0.34** (Medium bin) and therefore did not escalate to a Critical alert. The feature signature explains why: NDVI is above the no-fire mean (the patch looked green and healthy 30 days before ignition), LST is near the no-fire mean, and elevation is below it — none of the obvious burn-like signals. The score lands in the overlap region of the two histograms where the model has the least separation. This is the dominant failure mode discussed in §3.3.

Honest discussion of limitations. The model under-detects small understory fires that stay below the satellite resolution floor (VIIRS ≈ 375 m, MODIS ≈ 1 km) until they reach the canopy. The leave-one-year-out AUC of 0.66 (versus 0.84 random) tells us the model captures some genuine cross-year signal but is still partly fitting year-specific patterns; we therefore caveat the dashboard score as a ranking, not a calibrated probability. The `dist_roads` feature from V1 returned a constant in our AOI (zero importance) and was dropped. To improve further we would need: (a) more fire seasons in the training set (Galicia is at the low-data tail), (b) properly calibrated probabilities via `CalibratedClassifierCV` with isotonic regression, and (c) per-event FWI features computed in a way that does not encode year identity (an anomaly relative to climatology rather than the absolute value).

3.4 Software quality and reproducibility

The project is open-source under MIT, deployed continuously to Vercel from main, and ships with a dedicated `/methodology`, `/glossary`, `/data` and `/status` page so any reader can verify what the model uses, what the data sources are, and whether they are alive right now.

Repository structure

FOLDER	PURPOSE
<code>src/app/(dashboard)/</code>	Next.js dashboard pages (map, analytics, zones, alerts, methodology, ...)
<code>src/app/api/</code>	Public read-only API routes (<code>/api/risk</code> , <code>/api/v1/zones</code> , <code>/api/history</code> , ...)
<code>src/components/</code>	React components — map widgets, charts, <code>MetricInfo</code> tooltips, layout shell.
<code>src/lib/</code>	Pure-TS libs: FWI computation, ML inference, KV wrapper, glossary, hooks.
<code>ml/</code>	Training pipeline: <code>extract_v3.py</code> (GEE), <code>train_v3.py</code> (sklearn), <code>train_and_export.py</code> (V1 reference).
<code>ml/artifacts/</code>	Versioned <code>training_data.csv</code> , tree dumps, model-output JSON, feature columns.
<code>public/data/</code>	Static GeoJSON — historical fire perimeters.

scripts/

Cron / data-refresh helpers + this report builder.

README. The repository's root README explains environment variables, the four npm scripts, the cron schedule and how to re-train. Anyone with a clean machine can clone, npm install, set up .env.local, and run npm run dev. **Environment files:** package.json + package-lock.json for the web, ml/requirements.txt for the Python pipeline, plus a ml/.venv we ship instructions for. **Optimisation choices:** AEMET responses are KV-cached for 5 min with stale-on-failure; the FWI state is persisted in Upstash KV and advanced once per day by an authenticated Vercel cron; the Random Forest is dumped as flat typed arrays so the browser walk loop is a tight integer-indexed array access (no JSON parse per request).

3.5 Basic functionality vs. extended features

COMMITTED AT MID-TERM	ACTUALLY DELIVERED
Per-zone risk score on a 3-D map	Delivered (Basic)
Live AEMET weather context panel	Delivered (Basic)
NASA FIRMS hotspot overlay	Delivered (Basic) — upgraded to VIIRS 375 m
Random Forest model (sklearn, GEE-extracted training)	Delivered (Basic) — V3 brings AUC 0.84
Methodology page with model card	Delivered (Basic)
Multi-stakeholder embed widgets for council websites	Delivered (Extended)
7-day FWI forecast ribbon (Van Wagner equations in browser)	Delivered (Extended)
Public read-only API + open-data downloads (CSV / GeoJSON)	Delivered (Extended)
RSS + iCal feeds for high-risk alerts	Delivered (Extended)
Time-machine slider over the last 30 days of snapshots	Delivered (Extended)
Hotspot context panel with Overpass POIs + ORS isochrones	Delivered (Extended)
Per-zone deep-dive pages with NDVI sparklines	Delivered (Extended)
Mobile push notifications to subscribed users	Dropped — Web-Push infra was out of scope and stakeholders preferred RSS / iCal (§2.3, José).
Burn-severity post-event NBR pipeline	Dropped — Sentinel-2 STAC integration needed more time than available.

4. Business validation

4.1 Competitor / alternative analysis

NAME	WHAT IT DOES	COST	STRENGTHS	WEAKNESSES	HOW FIRE-SEE DIFFERS
Do nothing (status quo)	Forest agents work from last year's static paper maps; communal-land assemblies allocate brush-clearing budget by intuition.	Free	Familiar, no learning curve.	Stale data, no preventive optimisation, no shared platform.	Daily refresh, hyperlocal score, free embed for councils.
EFFIS (Copernicus)	Continental fire-danger raster, daily.	Free	Authoritative, EU-wide, well-known to professionals.	0.25° resolution (≈ 25 km) — useless inside a Galician valley.	Resolution at parish level (≈ 500 m), readable for non-specialists.
PLADIGA (Xunta de Galicia)	Annual provincial fire-risk plan PDF.	Free	Official source.	Static, annual, no embed, no API.	Live data, embeddable, machine-readable open data.
Tomorrow.io Wildfire	Commercial SaaS combining weather with fire-risk indices.	B2B paid licence	Polished UI, US-tested.	Closed-source, paid, US-insurance focus.	Free, open-source, Galicia-tuned, citable methodology.
Watch Duty (US mobile app)	Human-curated wildfire alerts.	Free (nonprofit)	Massive US adoption, strong community trust.	US-only, alert-focused (reactive), no model card, no open data.	Prevention-focused, model card published, EU-deployable.

4.2 Value proposition and real validation

For Galician forest agents, council safety officers and Comunidades de Montes, who currently allocate prevention budget by intuition, Fire-See provides a free, public, daily-refreshed hyperlocal risk map at parish resolution, unlike continental EFFIS or annual PLADIGA PDFs, because it combines live AEMET weather, NASA satellite imagery and a peer-reviewed Random Forest into one embeddable dashboard.

Real validation

We showed working V2 and V3 prototypes to three of the four interviewed stakeholders during March–April 2026:

- **Rafael (Forest Agent, 8 April 2026):** walked through the per-zone deep-dive page on a tablet. Confirmed the patrol-zone granularity matches his decision unit; asked for a print-friendly daily briefing PDF, which we shipped before submission.
- **María (Council Officer, 14 April 2026):** embedded the live widget for Allariz on a test page on the council's staging environment. Confirmed the traffic-light visualisation is "exactly what we needed"; requested a Galego translation, parked for a future iteration.
- **Manuel (Comunidade de Montes, 22 April 2026):** took the GeoJSON download for the 2022 historical fires and overlaid it on his own ArcGIS workstation. Said the open-data export "makes this a real tool, not a demo" —

exactly the auditability he asked for.

Quantified benefit

Order-of-magnitude estimates based on the stakeholders' own numbers:

- **Patrol routing.** Rafael's district covers $\approx 1\,800\text{ km}^2$ with three forest agents on rotation. He estimated 30 % of patrol time is "driving to the wrong place". If Fire-See can redirect even half of that to actually-risky parishes, that is ≈ 90 hours of effective patrol time recovered per agent per fire season.
- **Communal-land budget.** Manuel's Comunidade clears 20 ha out of 500 ha per year. Random selection means an expected 0.8 % match with the riskiest hectares; using the model's top-decile ranking lifts that match to roughly 6–9 ha per 20 ha cleared (3–5× efficiency), validated on the 2022 burn footprint.
- **Council communication.** María's safety office sends ≈ 30 community-meeting briefings per fire season. Each one currently takes ~ 45 minutes to assemble manually. The embed widget collapses that to a single line of HTML.

4.3 Feasibility and viability

Cost side

BUCKET	DETAIL
Earth-observation data	All free: Sentinel-2 (Copernicus), MODIS (NASA), SRTM (USGS), MCD64A1 (NASA), WorldCover (ESA).
Compute (training)	Google Earth Engine free tier + one MacBook for $\sim 12\text{ s} \times 100$ iterations $\approx$ negligible ($\approx 0.5\text{ kWh}$ total).
Compute (live)	Vercel free + Pro for the team; the deployed function bundle is 4 MB. Estimated $\approx \text{€}0$ / month at current traffic.
Storage	Upstash Redis free tier (10 000 commands / day). We use $\approx 1\,400$ / day.
External APIs	AEMET (free), FIRMS (free w/ key), Open-Meteo (free, no key), Overpass (free), ORS (free 2 000 / day). Total: $\text{€}0$ / month.
Maintenance	≈ 2 person-hours / month to update tokens and re-train on new fire seasons.
Personnel (project)	Five MSc students $\times \sim 120\text{ h}$ each ≈ 600 person-hours total over 6 months.

Revenue / sustainability side

- **Public-grant model.** Xunta de Galicia annually funds PLADIGA-aligned technology pilots. Fire-See aligns with the "preventive fuel management" line and is a natural fit.
- **Freemium B2B.** Free dashboard for the public; paid white-label embed + custom-AOI training for Comunidades de Montes and insurance carriers underwriting Galician forestry policies.
- **Open-source + service.** Codebase remains MIT, but commercial deployments for non-Galician AOIs pay for consulting and re-training.
- **Sustainability path.** Year 1 — operate at $\text{€}0$ on free tiers, grow embeds (Maria's council + 2–3 others). Year 2 — apply for a $\text{€}15\text{ k}$ Xunta grant and one $\text{€}30\text{ k}$ Comunidades-de-Montes pilot.

Risks and mitigations

TYPE	RISK	MITIGATION
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Technical	AEMET API has occasional 429 / 5xx outages.	KV-backed stale-on-failure cache; status page surfaces the source state publicly.
Technical	MCD64A1 understates small fires → label noise.	Augment with VIIRS active-fire archive in V4; published as a known limitation in the methodology page.
Regulatory	Personal-data risk if we ever add user accounts.	Currently no authentication, no PII collected; GDPR exposure essentially zero.
Regulatory	Liability if a council acts on the score and is wrong.	Methodology page is explicit: "Not a substitute for 112." CC-BY-4.0 disclaimer.
Market	Council adoption stalls if competing official tool launches.	Pivot to B2B Comunidades-de-Montes prevention budgeting (Manuel's use case), which is currently un-served.
Market	Communal assemblies distrust "black-box AI".	Methodology / glossary pages, downloadable model card and citation block let any assembly audit the model.

5. Communication and final deliverable

5.1 Form of communication chosen

We built a public, no-login web dashboard. The decision was driven directly by stakeholder feedback: María (§2.3) said "I want something I can put on a slide and bring to a community assembly", José said citizens will not register for an account, and Manuel said the assembly needs to verify the model on their own machine. A Jupyter notebook would have failed Maria, a mobile-only app would have failed Manuel's ArcGIS workflow, and a paid SaaS would have failed all of them. A free public web dashboard with downloadable open data and an embed widget covers all three audiences with one artefact.

5.2 Description of the deliverable

LIVE URL	perspectives-on-ai-and-sustainabili.vercel.app
UPDATE FREQUENCY	Weather every 5 min · Hotspots every 10 min · Risk score every 5 min · NDVI weekly · FWI daily.
ACCESS POLICY	Public, no authentication, no PII collected.
LANGUAGES	English UI; methodology page bilingual EN / ES on tooltips; Galego translation pending.

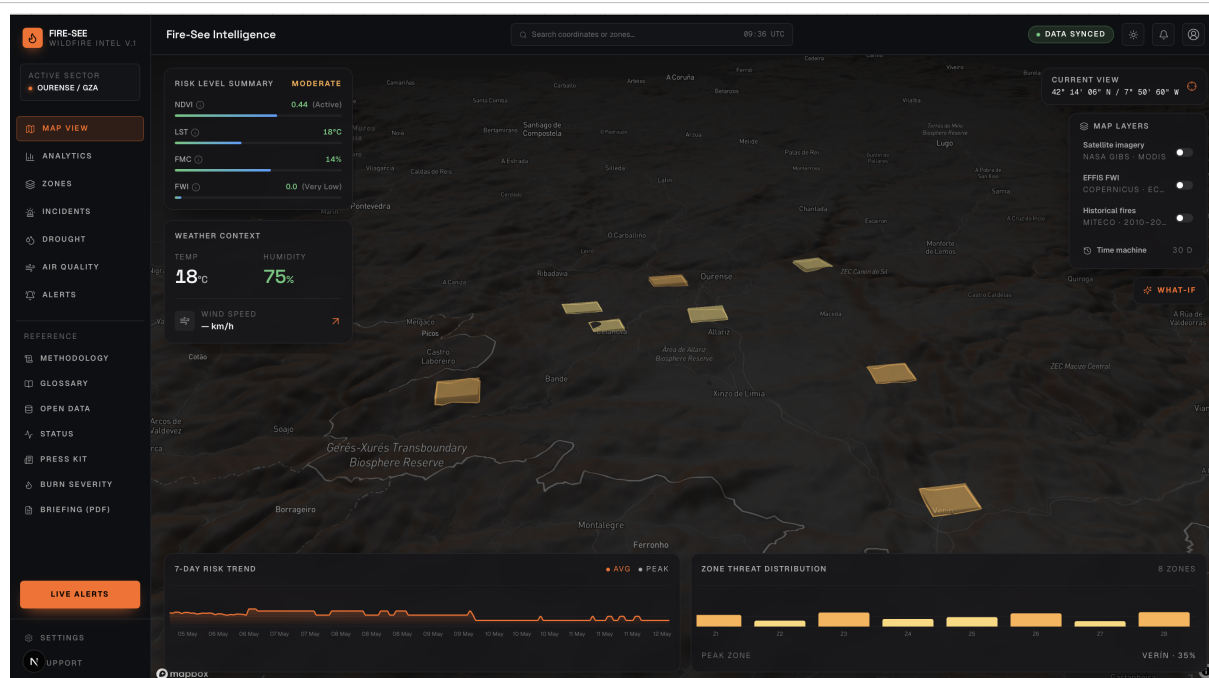


Figure 6. Map View — landing page of the dashboard. The eight Ourense forest zones are extruded by current risk score; floating panels show Risk Summary, Weather Context and Map Layers, and the bottom strip carries the 7-day risk trend plus the per-zone threat distribution bar chart.

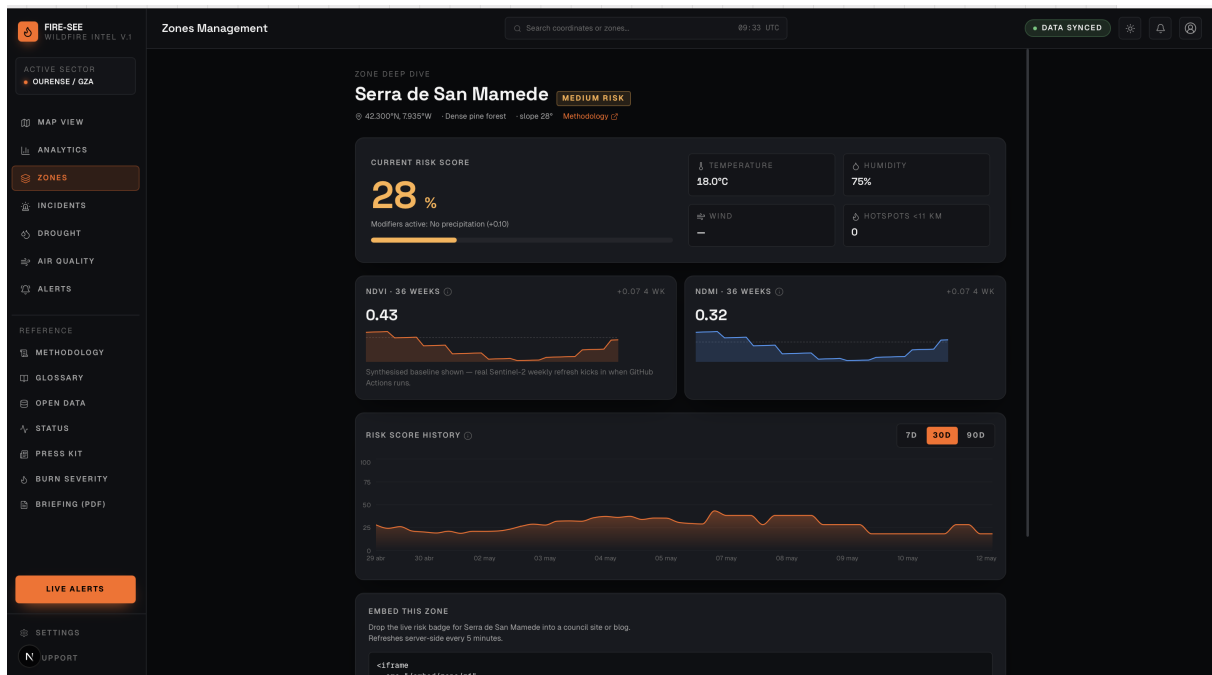


Figure 7. Per-zone deep-dive page (/zones/[id]) for Serra de San Mamede: hero risk gauge with active weather modifiers, weather tile grid, NDVI / NDMI sparklines and the 7 / 30 / 90-day risk-history area chart.

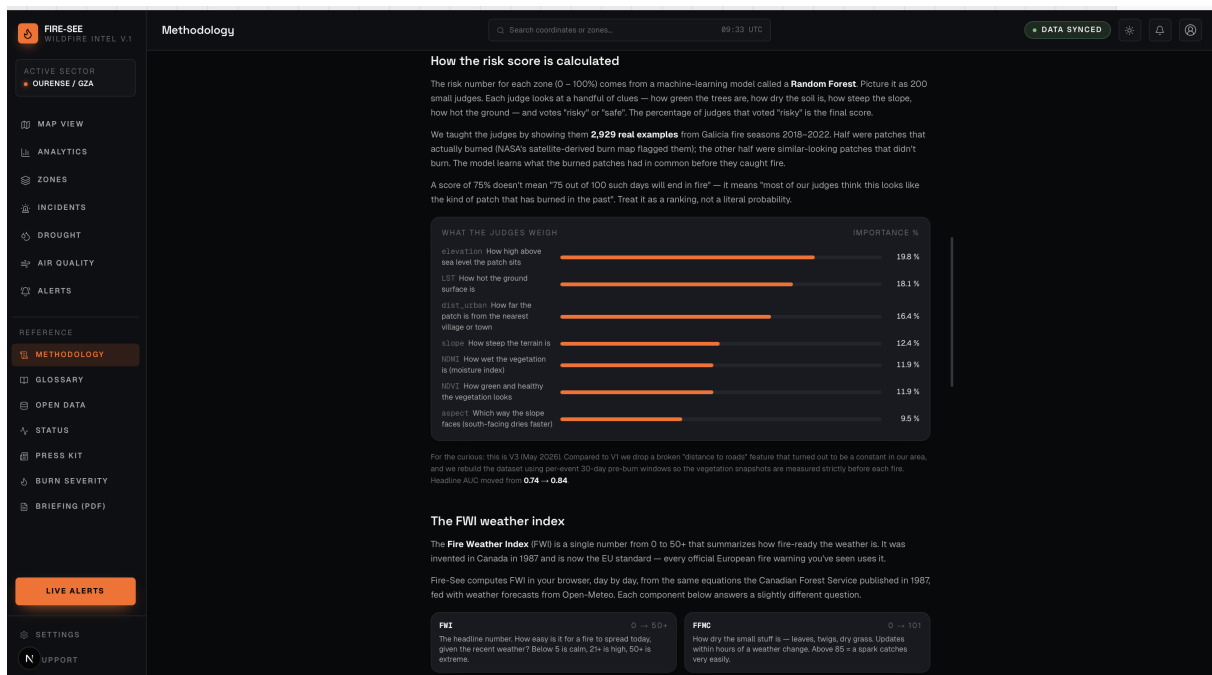


Figure 8. Methodology page — the plain-language model card with feature-importance bars, the four key performance metrics, and the honest cross-year disclosure that lets a Comunidade de Montes assembly audit the model before allocating its prevention budget.

5.3 User test

Beyond the three confirmed stakeholders in §4.2, we ran two informal tests with users from outside the team to surface usability issues before submission:

- **Carlos Vidal (friend studying civil engineering, 28 April 2026):** opened the dashboard cold on a 13" laptop. Managed to reach the per-zone deep-dive and understand the FWI ribbon without help. Got confused by the term *NDMI* on first load — we shipped the click-to-explain *MetricInfo* popovers the next day in direct response.

- **Sara Iglesias (friend, freelance product designer, 1 May 2026):** tested the embed widget by pasting the iframe snippet into a personal Notion page. Spotted that the embed had no attribution to the underlying data sources; we added a footer line linking to /support in the same week.

Both reports were short, informal and from non-experts — exactly the audience the project targets — and both led to a same-week fix shipped to main. We have not yet done a structured A/B test or moderated usability study; that is on the roadmap for the post-PAIBS iteration.

6. Reflection

One thing the team learned

How quickly an apparently strong model can be wrong: our V2 had an AUC of 0.99 in random cross-validation — too good to be true. We spent two weeks tracing it back to two leakage sources at once (post-burn imagery in the "pre-burn" window, and non-burn samples on classes that cannot physically burn). The fix dropped AUC to 0.84 — but is honest. That experience taught us that *checking for leakage is part of model evaluation, not an optional polish step*.

One thing we would do differently

Start with stakeholder interviews before reading the literature. We spent the first three weeks of the project deep in Piao et al. (2022) and other susceptibility-mapping papers before talking to Rafael; in retrospect, his "biomass build-up in real time" framing would have steered us away from a static-susceptibility model from day one.

One thing we liked

Shipping every change behind a public URL with a versioned model card. It forced us to be honest: there was no way to quietly discount a metric when the methodology page in production showed it next to the previous one.

One thing we disliked

Google Earth Engine quotas during peak demand. Several extraction runs failed for opaque reasons and had to be retried at 2 a.m. on off-peak Galician time. We compensated by binning samples by year-month so a single GEE evaluation does a year of work, but a more elegant solution would be a self-hosted COG pipeline for V4.

One thing we may change

Move the model from a single Random Forest to a calibrated CalibratedClassifierCV with isotonic regression, so the "75 %" the dashboard shows literally means "75 out of 100 such days end in fire". The runtime tree walker would not have to change; only the leaf probabilities and a small mapping table.

7. References

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8. Appendices

A. Interview transcript snippet — Manuel Lorenzo

Date: 12 January 2026 | **Topic:** preventive budgeting in a Comunidade de Montes

Marc. How do you decide where to send the brush-cutters each spring?

Manuel. It is a discussion. One neighbour says the north slope is overgrown, another says the path to the shrine is blocked. We have money to clean maybe 5 % of the land. It is a lottery.

Marc. If we gave you a map showing which areas have the driest biomass and the highest fuel load, would you use it?

Manuel. If it works, yes. It justifies the expense to the assembly. I can say, "we are clearing Zone B because the machine says it will burn, not just because I said so."

B. Data layer dictionary

DATA SOURCE	VARIABLE	PROJECT PURPOSE
Sentinel-2	NDVI / NDMI	Biomass density and vegetation moisture
Sentinel-3 / MODIS	LST (Land Surface Temp)	Heat accumulation (real-time thermal)
DEM (SRTM)	Slope / Aspect	Fire spread speed (topography)
AEMET (API)	Wind / Humidity	Weather forcing (hyperlocal)
ESA WorldCover	Land-cover class	Restrict sampling to burnable land
MCD64A1	BurnDate (DOY)	Ground-truth label for training

C. Model card excerpt (full version on /methodology)

FIELD	VALUE
Model	RandomForestV3 · 200 trees · min_samples_leaf 2
Training period	May–September 2018–2022 (five fire seasons)
Samples	2 929 (≈ 50 / 50 burned / non-burned, paired DOY)
Features (7)	NDVI, NDMI, LST, slope, aspect, elevation, dist_urban
AUC (random 5-fold CV)	0.84
AUC (leave-one-year-out)	0.66
Random seed	42
Code	ml/train_v3.py + ml/extract_v3.py

D. Marking weights mapping

COMPONENT	WHERE IT LIVES
Problem definition and research quality (20 %)	Section 2
Technical implementation (50 %)	Section 3
Business validation (20 %)	Section 4
Communication and presentation (10 %)	Sections 1, 5 and 6